

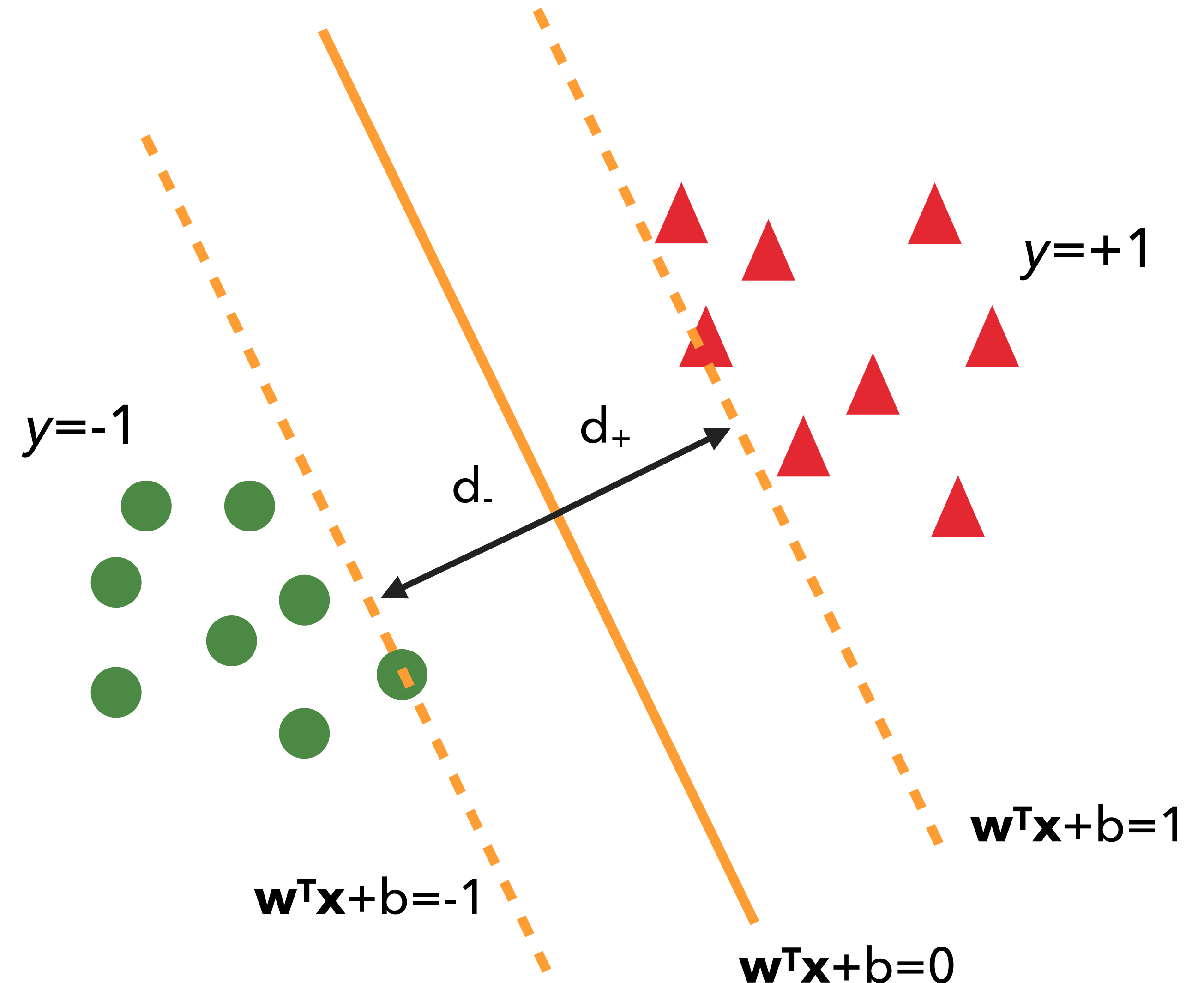
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# DATA MINING

# SVM: KNOWLEDGE REPRESENTATION AND SCORING FUNCTION

- ▶ Linear SVM:  $y = \text{sign} \left[ \sum_{i=1}^m w_i x_i + b \right]$
- ▶ Margin =  $d_+ + d_- = 2/\|\mathbf{w}\|$
- ▶ Optimization problem
  - ▶  $\max 2/\|\mathbf{w}\|$
  - ▶ subject to  
 $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i \in \{1, 2, \dots, N\}$



## SVM LEARNING

- ▶ Equivalent to minimize  $\|\mathbf{w}\|^2/2$  subject to

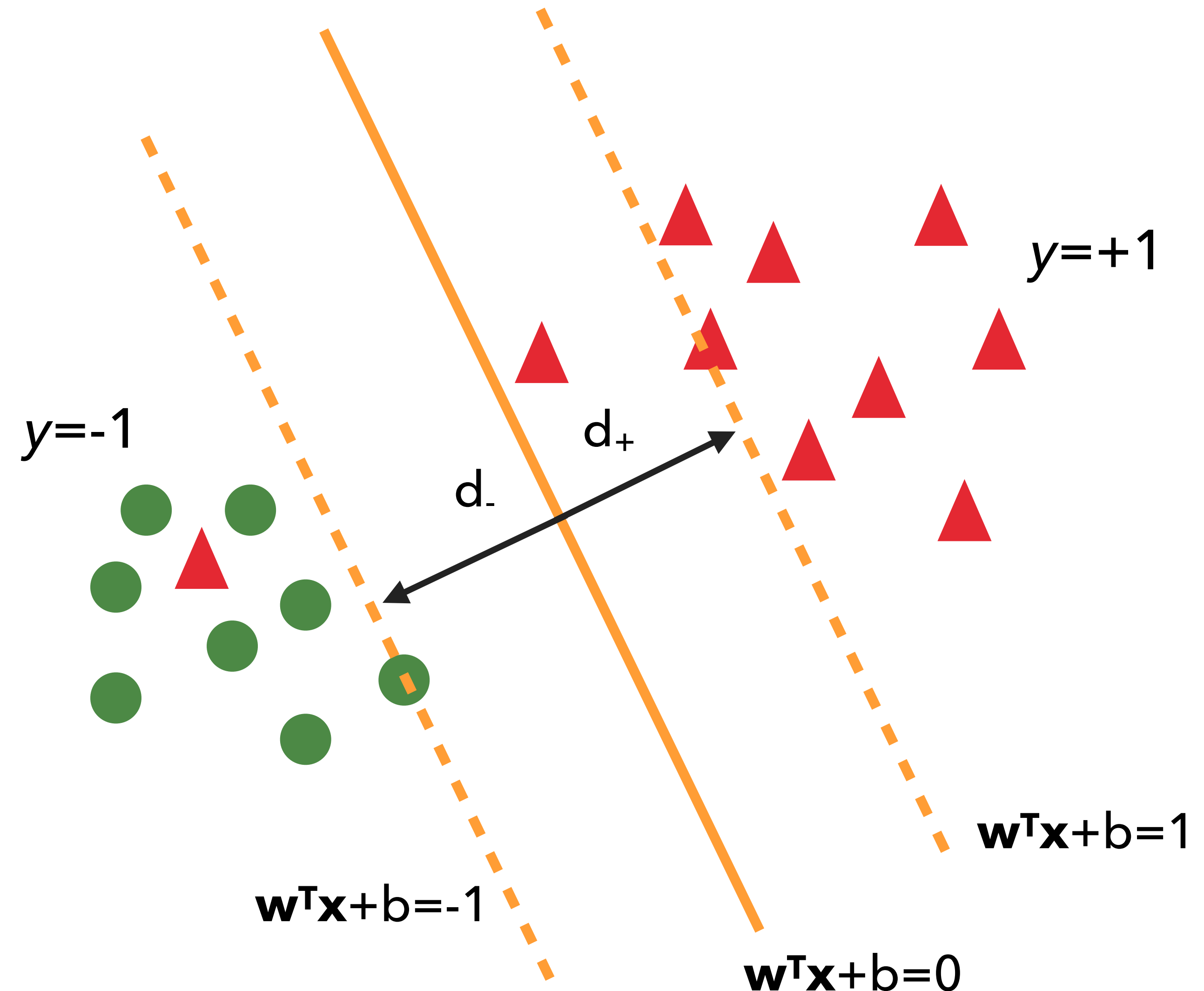
$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i \in \{1, 2, \dots, N\}$$

- ▶ This is a **quadratic optimization** problem subject to linear constraints, there is a unique minimum

- ▶ Lagrangian function  $L(\mathbf{w}, b, \lambda_i) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_{i=1}^N \lambda_i(1 - y_i(\mathbf{w}^T \mathbf{x}_i + b))$

## WHAT ABOUT LINEARLY NON-SEPARABLE DATA?

- ▶ Introduce slack variables  $\varepsilon_i \geq 0$  such that:  
$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \varepsilon_i, \forall i \in \{1, 2, \dots, N\}$$
- ▶  $\varepsilon_i$  measures the amount of error
  - ▶ When  $0 < \varepsilon_i \leq 1$ , data is between the margin, but classified correctly
  - ▶ When  $\varepsilon_i > 1$ , data is misclassified



## “SOFT” MARGIN OPTIMIZATION

- ▶ With slack variables the score function is:

$$\min_{\mathbf{w}, \xi} \|\mathbf{w}\|^2 + C \sum_i^N \xi_i$$

- ▶ And new constraints:

$$y_i(x_i \cdot w + b) - (1 - \xi_i) \geq 0 \quad \forall i$$

- ▶ If  $\xi$  are sufficiently large, then every constraint can be satisfied
- ▶ C is regularization parameter
  - ▶ Small C means constraints can be ignored in order to find large margin
  - ▶ Large C means constraints cannot be ignored and result is small margin (C= $\infty$  enforces hard margin)

## SVM OPTIMIZATION

- ▶ Constraint can be rewritten as:

$$y_i f(x_i) \geq 1 - \xi_i \quad \forall i$$

- ▶ Together with  $\xi_i \geq 0$ , is equivalent to:

$$\xi_i = \max\left(0, 1 - y_i f(x_i)\right)$$

- ▶ Hence we can use the following score in unconstrained optimization:

$$\min_{\mathbf{w}} \|\mathbf{w}\|^2 + C \sum_i^N \left[ \max\left(0, 1 - y_i f(x_i)\right) \right]$$

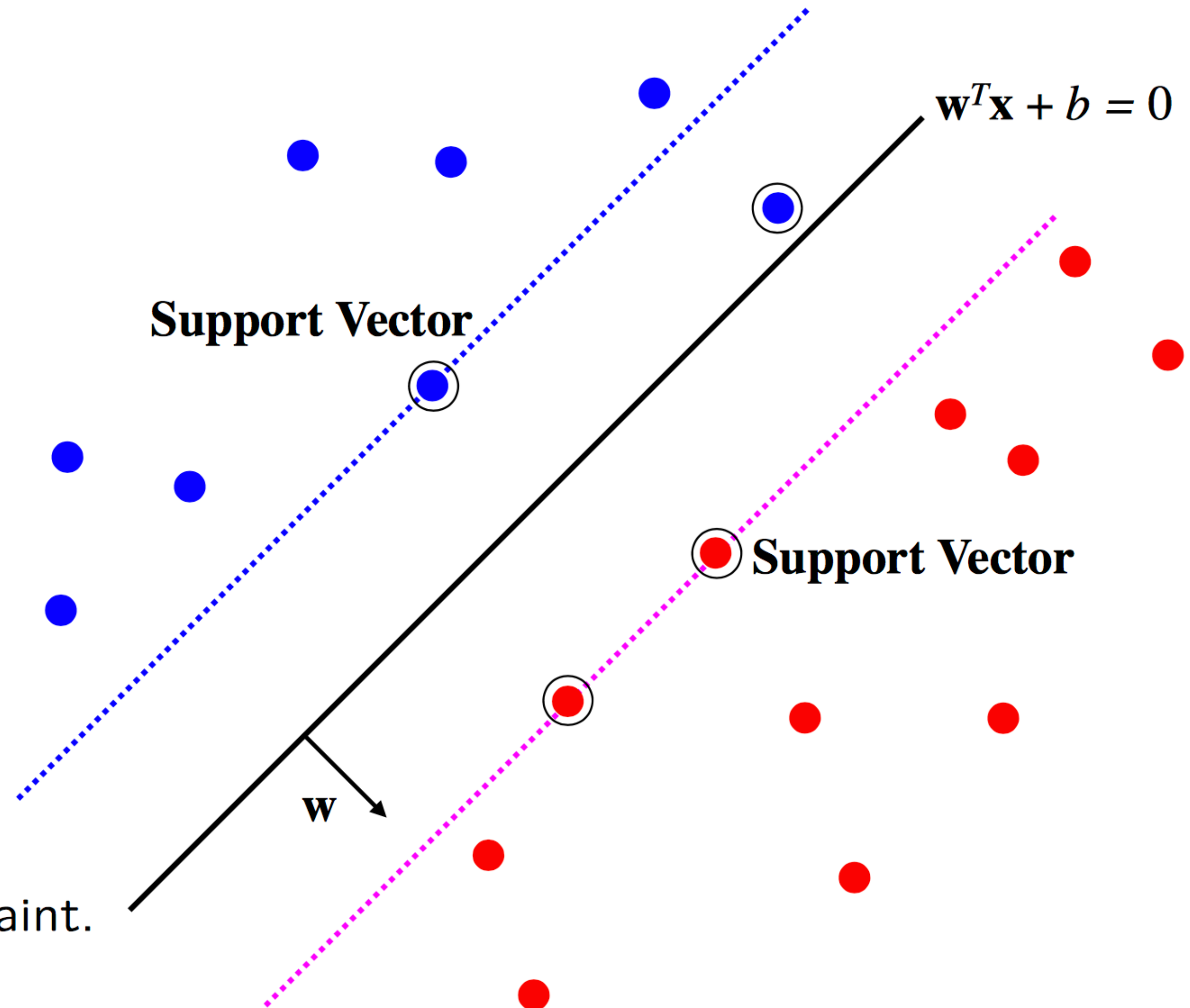
## NEW OBJECTIVE

$$\min_{\mathbf{w}} \|\mathbf{w}\|^2 + C \sum_i^N \left[ \max\left(0, 1 - y_i f(x_i)\right) \right]$$

**Hinge Loss**

Points are in three categories:

1.  $y_i f(x_i) > 1$   
Point is outside margin.  
No contribution to loss
2.  $y_i f(x_i) = 1$   
Point is on margin.  
No contribution to loss.  
As in hard margin case.
3.  $y_i f(x_i) < 1$   
Point violates margin constraint.  
Contributes to loss





## REWRITE THE OBJECTIVE FUNCTION

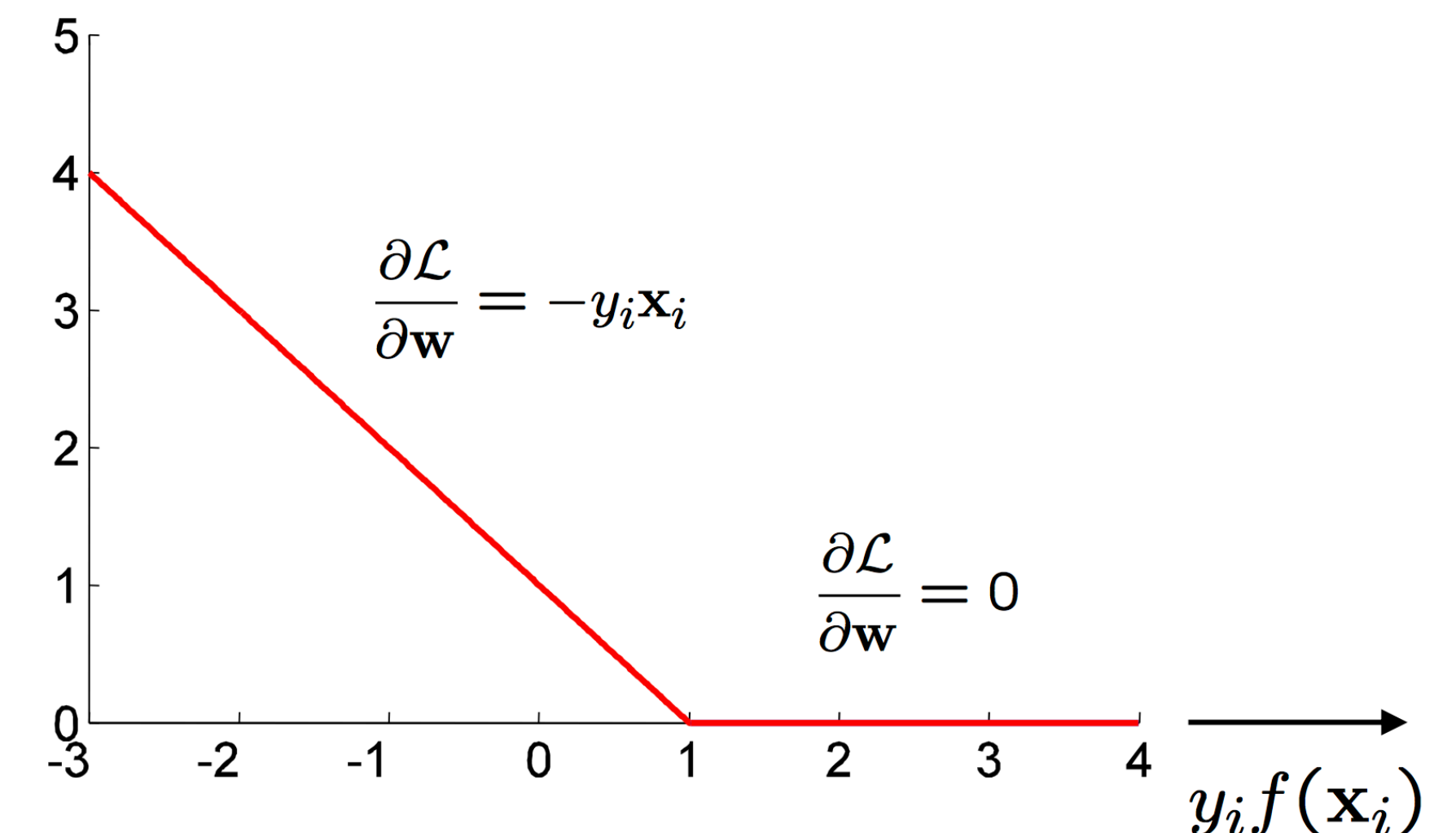
$$\min_{\mathbf{w}} ||\mathbf{w}'||^2 + C \sum_i^N \left[ \max \left( 0, 1 - y_i f(x_i) \right) \right]$$

# SVM OPTIMIZATION WITH SUB-GRADIENT

- ▶ Rewrite optimization problem:

$$\min_{\mathbf{w}} \frac{1}{N} \sum_i \left( \frac{\lambda}{2} \|\mathbf{w}\|^2 + \left[ \max \left( 0, 1 - y_i f(x_i) \right) \right] \right)$$

- ▶ Now  $\lambda = \frac{2}{N \cdot C}$ , becomes the regularization parameter
- ▶ Hinge loss is not differentiable however—so must use sub-gradient for optimization



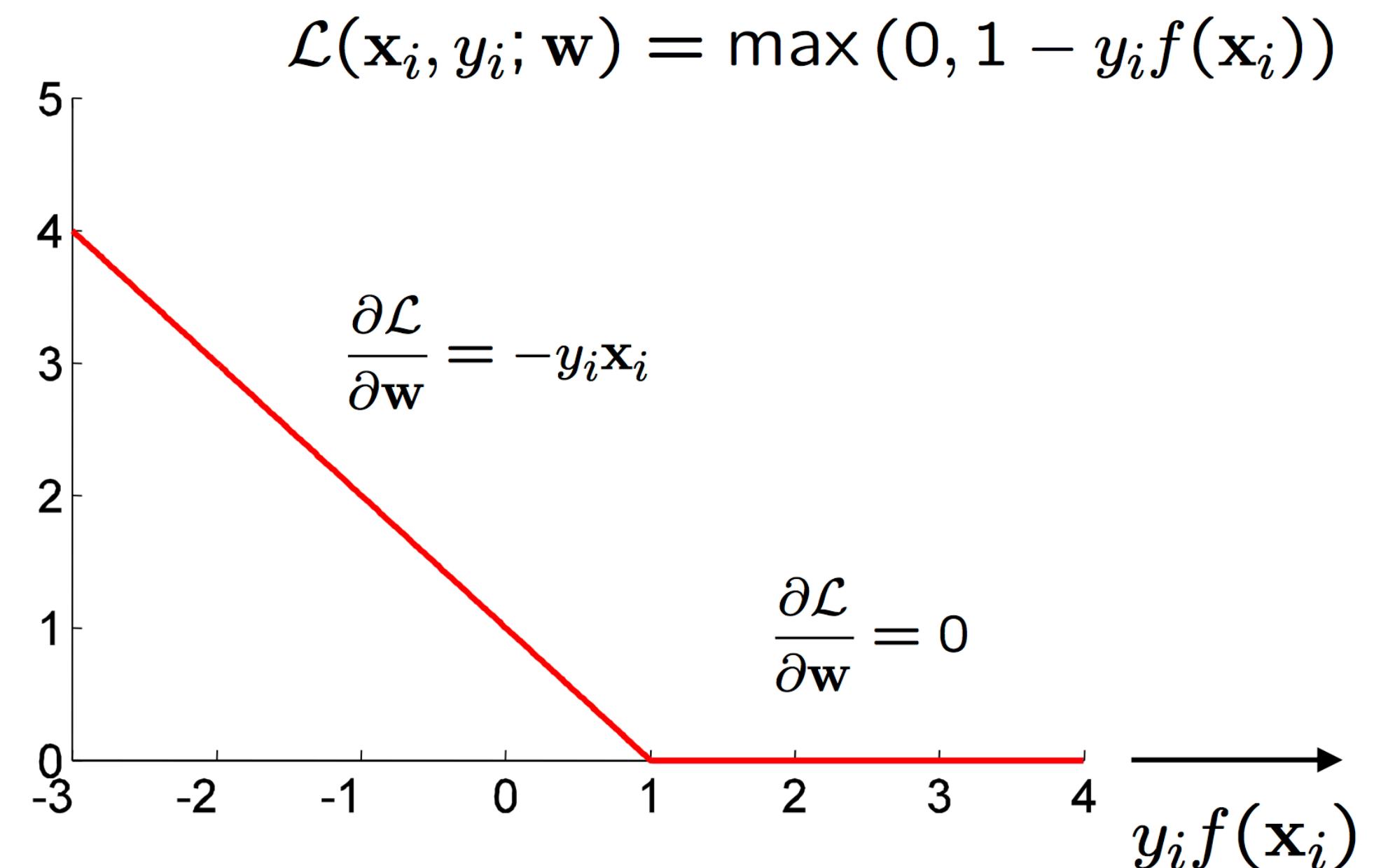
## SUB-GRADIENT DESCENT

- ▶ Iterative update for SVM weights using sub-gradient descent:

$$\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta \frac{1}{N} \sum_i^N (\lambda \mathbf{w}_t + \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{x}_i, y_i; \mathbf{w}_t))$$

- ▶ where  $\eta$  is the learning rate as per usual
- ▶ Each iteration cycles through the data:

$$\begin{aligned} \mathbf{w}_{t+1} &\leftarrow \mathbf{w}_t - \frac{\eta}{N} (\lambda \mathbf{w}_t - y_i \mathbf{x}_i) && \text{if } y_i f(\mathbf{x}_i) < 1 \\ &\leftarrow \mathbf{w}_t - \frac{\eta}{N} \lambda \mathbf{w}_t && \text{otherwise} \end{aligned}$$



# SVM EXAMPLE

| Intercept | Age>40 | Income=high | Student=yes | Credit=fair | BuysComp? |
|-----------|--------|-------------|-------------|-------------|-----------|
| 1         | 0      | 1           | 0           | 1           | -1        |
| 1         | 0      | 1           | 0           | 0           | -1        |
| 1         | 0      | 1           | 0           | 1           | +1        |
| 1         | 1      | 0           | 0           | 1           | +1        |
| 1         | 1      | 0           | 1           | 1           | +1        |
| 1         | 1      | 0           | 1           | 0           | -1        |
| 1         | 0      | 0           | 1           | 0           | +1        |
| 1         | 0      | 0           | 0           | 1           | -1        |
| 1         | 0      | 0           | 1           | 1           | +1        |
| 1         | 1      | 0           | 1           | 1           | +1        |
| 1         | 0      | 0           | 1           | 0           | +1        |
| 1         | 0      | 0           | 0           | 0           | +1        |
| 1         | 0      | 1           | 1           | 1           | +1        |
| 1         | 1      | 0           | 0           | 0           | -1        |

$BC = +1 \quad \text{if } [\mathbf{w}^T \mathbf{x}] > 0$   
 $BC = -1 \quad \text{otherwise}$

$\mathbf{x} = [Int, A, I, S, CR]$   
 $\mathbf{w} = [w_0, w_A, w_I, w_S, w_{CR}]$

**SVM parameters = w**

- ▶ Score function: margin + hinge loss on errors
- ▶ Estimate **w** to maximize margin, while minimizing errors

# SVM LEARNING

- Score function: soft margin (includes hinge loss on errors)

$$\min_{\mathbf{w}} \frac{1}{N} \sum_i^N \left( \frac{\lambda}{2} \|\mathbf{w}\|^2 + \left[ \max \left( 0, 1 - y_i f(x_i) \right) \right] \right)$$

- Estimate  $\mathbf{w}$  by minimizing objective function using gradient descent

## Gradient descent:

Start at some  $\mathbf{w}$ , e.g.,  $\mathbf{w}=[0,0,0,0,0]$

Make predictions given current  $\mathbf{w}$ :

$$\forall i \quad \hat{y}_i = \mathbf{w}^T \mathbf{x}_i$$

Calculate gradient for each parameter:

$$\forall j \quad \nabla_j = \frac{1}{N} \left[ \sum_{i=1}^n (\lambda w_j - \nabla_{ji}) \right]$$

where  $\nabla_{ji} = y_i x_{ij}$  if  $y_i \hat{y}_i < 1$ ; 0 otherwise

Move parameters in direction of gradient:  $\forall j \quad w_j^{new} = w_j - \eta \nabla_j$

Repeat until stopping criteria is met

# SVM PREDICTION

► What is the probability that new person will buy a computer?

| Intercept | Age>40 | Income=high | Student=yes | Credit=fair | BuysComp? |
|-----------|--------|-------------|-------------|-------------|-----------|
| 1         | 0      | 1           | 0           | 1           | -1        |
| 1         | 0      | 1           | 0           | 0           | -1        |
| 1         | 0      | 1           | 0           | 1           | +1        |
| 1         | 1      | 0           | 0           | 1           | +1        |
| 1         | 1      | 0           | 1           | 1           | +1        |
| 1         | 1      | 0           | 1           | 0           | -1        |
| 1         | 0      | 0           | 1           | 0           | +1        |
| 1         | 0      | 0           | 0           | 1           | -1        |
| 1         | 0      | 0           | 1           | 1           | +1        |
| 1         | 1      | 0           | 1           | 1           | +1        |
| 1         | 0      | 0           | 1           | 0           | +1        |
| 1         | 0      | 0           | 0           | 0           | +1        |
| 1         | 0      | 1           | 1           | 1           | +1        |
| 1         | 1      | 0           | 0           | 0           | -1        |
| 1         | 0      | 1           | 0           | 0           | ?         |

$\mathbf{x} = [1, 0, 1, 0, 0]$   
 $\mathbf{w} = [-.5, 1.2, 3, -2, 0.7]$   
 $\mathbf{x}^T \mathbf{w} = 2.5$   
 $BC = +1$